

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1146

Lake Goodwin 138 kV
January 25, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1146. The Interconnection Customer proposes a 120.2 MW wind facility interconnecting at the Lake Goodwin 138 kV substation in the Pennsylvania area. The request was received on February 09, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 299 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,785,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LAKE GOODWIN-ALLEGHENY NO. 5 HYDRO STA 138 KV	SUGAR HILL-ALLEGHENY ENERGY UNITS 3 4 & 5 345 KV	149.5	217	55.7
ALLEGHENY NO. HYDRO STA-STINSON 138 KV	SUGAR HILL-ALLEGHENY ENERGY UNITS 3 4 & 5 345 KV	143.7	222	55.7
SOUTHEAST VICKSBURG 345/138 kV Transformer Bank	ALLEGHENY ENERGY UNITS 3 4 & 5-SOUTHEAST VICKSBURG 500 KV	133.0	376	10.8
SLOSS INDUSTRIES CORP-REGAL 138 KV	ALLEGHENY NO. 5 HYDRO STATION-LAKE GOODWIN 138 KV	125.6	220	48.7
SLOSS INDUSTRIES CORP-SOUTHEAST VICKSBURG 138 KV	LAKE GOODWIN-ALLEGHENY NO. 5 HYDRO STA 138 KV	121.8	224	48.7
PLACER-STINSON 138 KV Line	ALLEGHENY ENERGY UNITS 3 4 & 5-SOUTHEAST VICKSBURG 500 KV	120.7	331	30.2
SUGAR HILL 345/138 kV Transformer	RITTER PARK-SUGAR HILL 345 KV	120.6	465	7.5
YUCCA-REGAL 138 KV	LAKE GOODWIN-ALLEGHENY NO. 5 HYDRO STA 138 KV	115.6	246	48.7
PACOLET DIESEL GENERATION FACILITY-SANDESTIN 138 KV	PLACER-STINSON 138 KV L	106.2	200	19.0
PLACER 138/1kV Autotransformer	ALLEGHENY ENERGY UNITS 3 4 & 5-SOUTHEAST VICKSBURG 500 KV	105.6	378	30.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LAKE GOODWIN-ALLEGHENY NO. 5 HYDRO STA 138 KV	Rebuild line with higher-capacity conductor	\$10,445,000	71% (\$7,395,000)
ALLEGHENY NO. HYDRO STA-STINSON 138 KV	Rebuild line with higher-capacity conductor	\$2,840,000	71% (\$2,010,000)
SOUTHEAST VICKSBURG 345-138 kV TX	Replace transformer with higher-rated unit	\$19,933,000	33% (\$6,567,000)
REGAL-SLOSS INDUSTRIES CORP SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$6,686,000	55% (\$3,707,000)
SLOSS INDUSTRIES CORP-SOUTHEAST VICKSBURG 138 KV Line	Rebuild line with higher-capacity conductor	\$4,261,000	55% (\$2,363,000)
PLACER-STINSON 138 KV Line	Rebuild line with higher-capacity conductor	\$1,652,000	64% (\$1,066,000)
SUGAR HILL 345/138 kV Transformer	Replace transformer with higher-rated unit	\$17,939,000	6% (\$1,102,000)
YUCCA-REGAL 138 KV	Rebuild line with higher-capacity conductor	\$1,934,000	55% (\$1,072,000)
SANDESTIN-PACOLET DIESEL GENERATION FACILITY 138 KV	Reconductor line	\$612,000	100% (\$612,000)
PLACER 138/1kV Autotransformer	Replace transformer with higher-rated unit	\$6,030,000	64% (\$3,891,000)
	Total allocated to project		\$29,785,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.